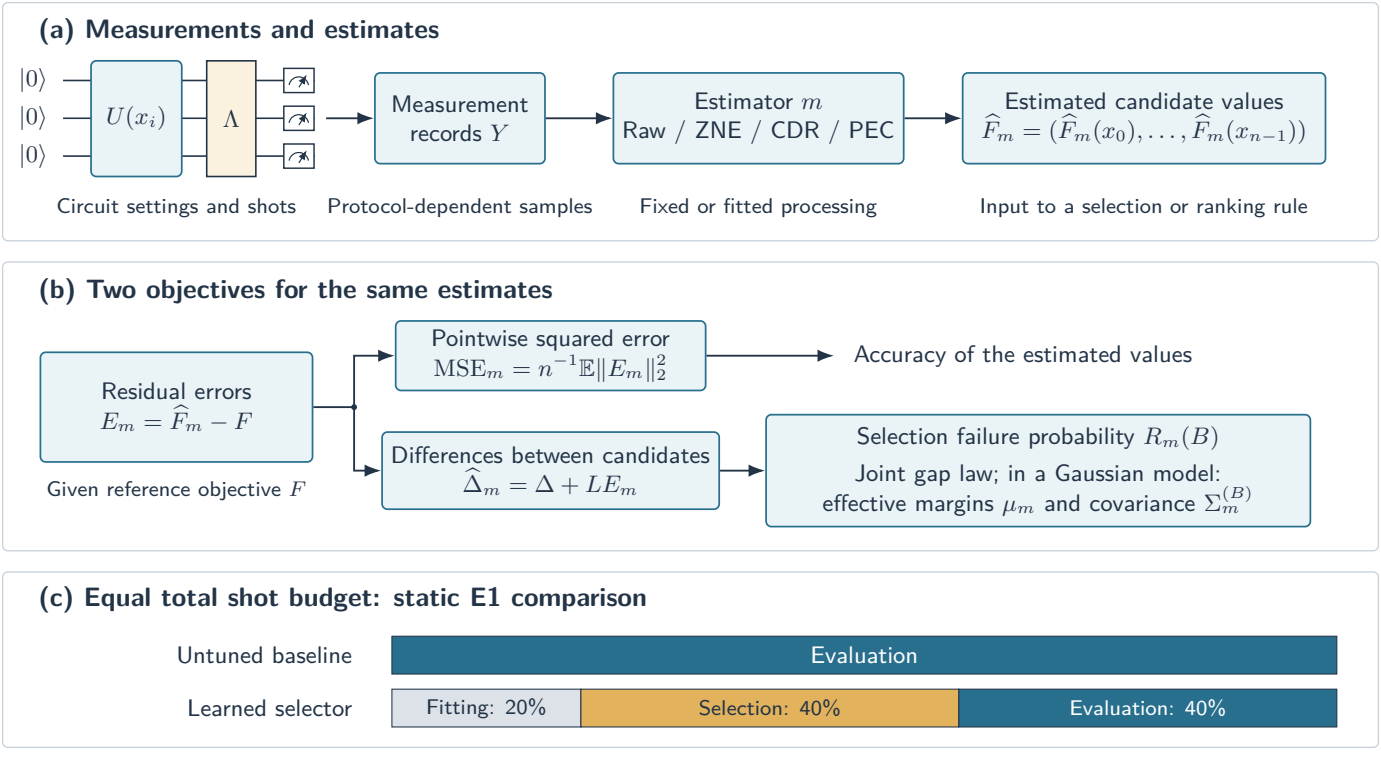

(a) Measurements and estimates
$|0\rangle$
$|0\rangle$
$|0\rangle$
$U(x_i)$
Λ
Measurement records Y
Estimator m
Raw / ZNE / CDR / PEC
Estimated candidate values
$\widehat{F}_m = (\widehat{F}_m(x_0), \ldots, \widehat{F}_m(x_{n-1}))$
Circuit settings and shots
Protocol-dependent samples
Fixed or fitted processing
Input to a selection or ranking rule
(b) Two objectives for the same estimates
Residual errors
$E_m = \widehat{F}_m - F$
Given reference objective F
Pointwise squared error
$\mathrm{MSE}_m = n^{-1}\mathbb{E}\|E_m\|_2^2$
Accuracy of the estimated values
Differences between candidates
$\widehat{\Delta}_m = \Delta + LE_m$
Selection failure probability $R_m(B)$
Joint gap law; in a Gaussian model:
effective margins μ_m and covariance $\Sigma_m^{(B)}$
(c) Equal total shot budget: static E1 comparison
Untuned baseline
Evaluation
Learned selector
Fitting: 20%
Selection: 40%
Evaluation: 40%